Prof. Dr Andreas Greiner (left) and Prof. Dr Seema Agarwal (right) with equipment for electrospinning at the University of Bayreuth. With backlighting, one can see the thin fibres from which nonwoven materials are formed (Photo credit: Christian Wißler)

The researchers achieved these materials through a special production process. In contrast to common methods of production, metal wires were not inserted into finished textiles. Rather, the scientists modified classical electro-spinning, which has been used to produce nonwovens for many years: Short electro-spun polymer fibres and small amounts of tiny silver wires with a diameter of only 80 nanometres are mixed in a liquid. Afterwards, these are filtered, dried and briefly heated up. If the composition is right, the resulting nonwoven material exhibits a very high degree of electrical conductivity.

New method to wake up devices

As smartphone users know all too well, a sleeping device can still suck the life out of a battery. One solution for extending the battery life of wireless devices under development by researchers at Stanford University is to add a wake-up receiver that can turn on a shut-off device at a moment's notice.

(R) Amin Arbabian, assistant professor of electrical engineering, and (L) graduate student Angad Rekhi demonstrate their ultrasonic wake-up receiver and the circuit boards used to test its performance (Image credit: Arbabian Lab)

Angad Rekhi, a graduate student in the Arbabian Lab at Stanford, and Amin Arbabian, assistant professor of electrical engineering, have developed a wake-up receiver that turns on a device in response to incoming ultrasonic signals—signals outside the range that humans can hear. By working at a significantly smaller wavelength and switching from radio waves to ultrasound, this receiver is much smaller than similar wake-up receivers that respond to radio signals, while operating at extremely low power and with extended range.

This wake-up receiver has many potential applications, particularly in designing the next generation of networked devices, including so-called smart devices that can communicate directly with one another without human intervention.

"As technology advances, people use it for applications that you could never have thought of. The Internet and the cellphone are two great examples of that," said Rekhi. "I'm excited to see how people will use wake-up receivers to enable the next generation of the Internet of Things."

Universal AI chip

China is standing out in the artificial intelligence (AI) race with the development of a system-on-chip that can add AI to any gadget. Developed by researchers from Beijing-based Tsinghua University, the Thinker chip is designed to support neural networks. Thus it could recognise objects in images and understand human speech. What makes the chip stand out is its ability to dynamically tailor its computing and memory requirements to meet the needs of the software being run. Also remarkable is the fact that a mere eight AA batteries are enough to power it for a year.

Thinker chip designed to handle AI tasks (Photo credit: Shouyi Yin, Tsinghua University Institute Of Microelectronics)

Low power requirements and flexible design make Thinker a unique offering in the AI field, as well as a major win for the Chinese tech sector. The chip's versatile design allows its implementation in everything from smartphones and computers to sensors and general offerings from the Internet of Things segment.